

Temperature & Humidity Sensor with RS485 (Modbus RTU)

1. Product Overview

WIN-SN-TNH-M-D is a versatile and reliable solution for anyone needing **precise temperature and humidity** monitoring with robust communication capabilities. It measures temperature and humidity in various environments, from industrial facilities to greenhouses and smart homes. This board stands out because it can communicate its readings through an isolated **RS485** interface. This type of communication is robust, making it ideal for long-distance data transmission and resistant to electrical interference, ensuring accurate and reliable measurements.



One of the key advantages of the WIN-SN-TNH-M-D is its flexibility in sensors compatibility. It supports a variety of sensors those are listed in Table 1 each suited for different range of temperature and humidity. Additionally, it features a 4-digit 7-segment display that provides real-time measurement data, allowing users to monitor sensor readings directly without requiring an external interface.

This manual provides instructions for installation and use of the sensor board to measure temperature and humidity in the environment. The sensor board offers data output in Modbus RTU (RS485) and can be powered by DC power supply.

	Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
☐	AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
☐	AHT20	-40°C ~ 85°C.	0~80% RH.	-40°C ~ 85°C.	0.3°C, 2.0 %RH
☐	SHT21	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.3°C, 2.0 %RH
☐	SHT25	-40°C ~ 125°C.	0~80% RH.	-40°C ~ 125°C.	0.2°C, 1.8 %RH

	Product Configuration
☐	TnH + RS485
☐	TnH + RS485 + Display
☐	TnH + WiFi
☐	TnH + WiFi + Display

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None

- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40001	0x03	RS485	16-bit int
2	Humidity	40002	0x03	RS485	16-bit int
1	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
2	Enter New Baud Rate	40012	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
4	Enter New Slave ID	40014	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
6	Enter New Parity	40016	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

5. Modbus RS485 Data Storage register

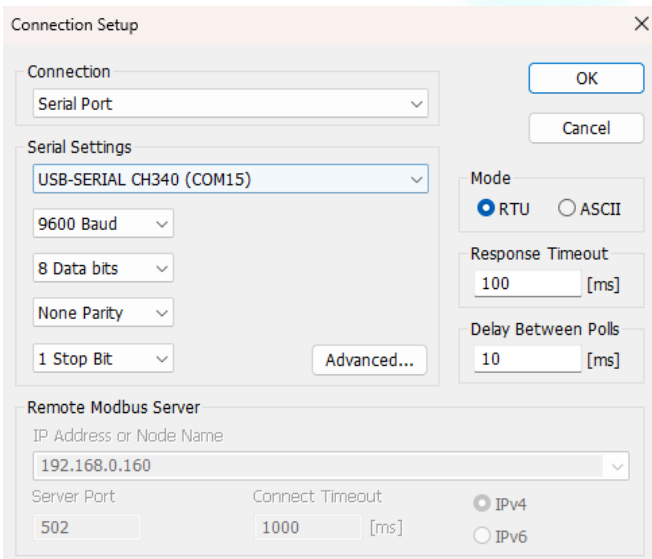
6. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.



Connection Setup

Connection: Serial Port

Serial Settings: USB-SERIAL CH340 (COM15)

Mode: ☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

Advanced...

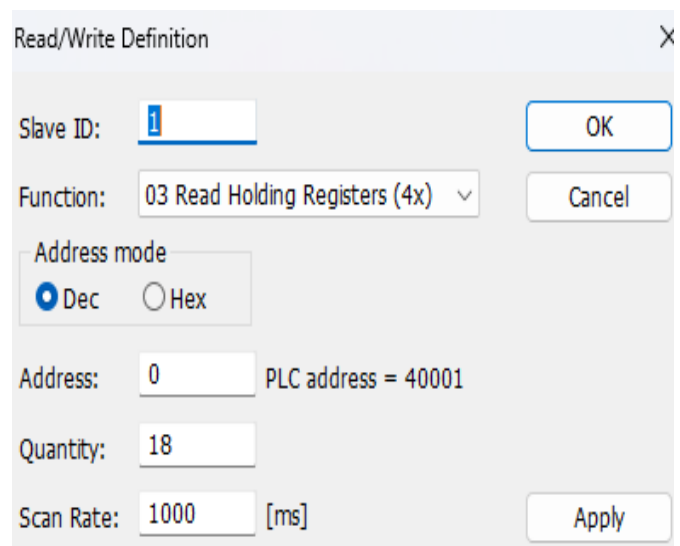
Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502

Connect Timeout: 1000 [ms]

☒ IPv4 ☐ IPv6



Read/Write Definition

Slave ID: 1

Function: 03 Read Holding Registers (4x)

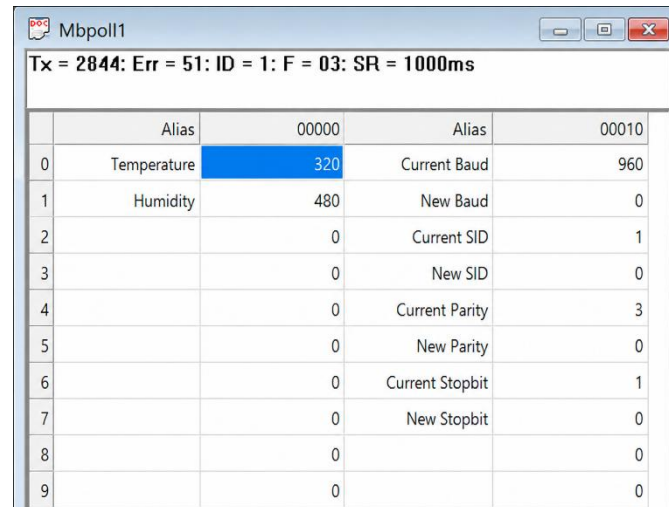
Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

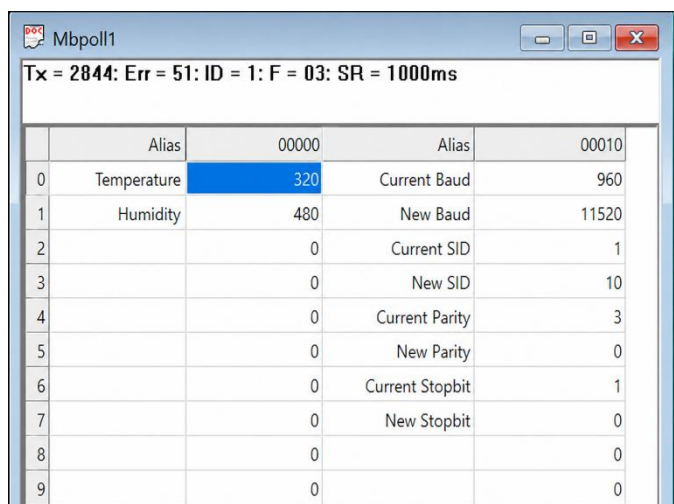
Scan Rate: 1000 [ms]

Apply

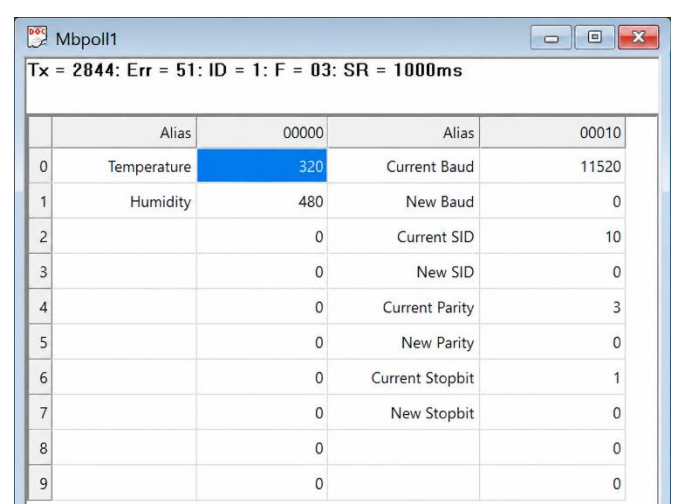


	Alias	00000
0	Temperature	320
1	Humidity	480
2		0
3		0
4		0
5		0
6		0
7		0
8		0
9		0

- Above image Shows sensor data over Modbus with default Modbus settings
- At the register 40001 shows Temperature Values, register 40002 shows Humidity Values.
- The value 320 indicates 32.0 temperature and 480 indicates as 48.0 humidity.
- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.



	Alias	00000
0	Temperature	320
1	Humidity	480
2		0
3		0
4		0
5		0
6		0
7		0
8		0
9		0



	Alias	00010
0	Temperature	320
1	Humidity	480
2		0
3		0
4		0
5		0
6		0
7		0
8		0
9		0

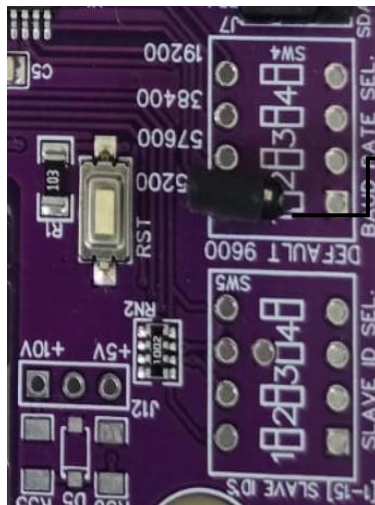


Image1.1



Image1.2

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

5. Device Configuration Process

1. Connect the Device

- Connect 12v-1A DC adaptor to power the board
- Connect the Wi-Fi SSID: WittelB-SN-TnH Pass:123456789
- Enter the IP 192.168.4.1 in web browser
- Refer image 1 for Wi-Fi WittelB-SN-TnH

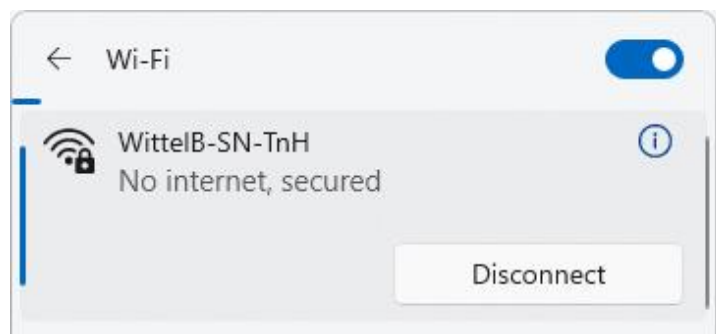
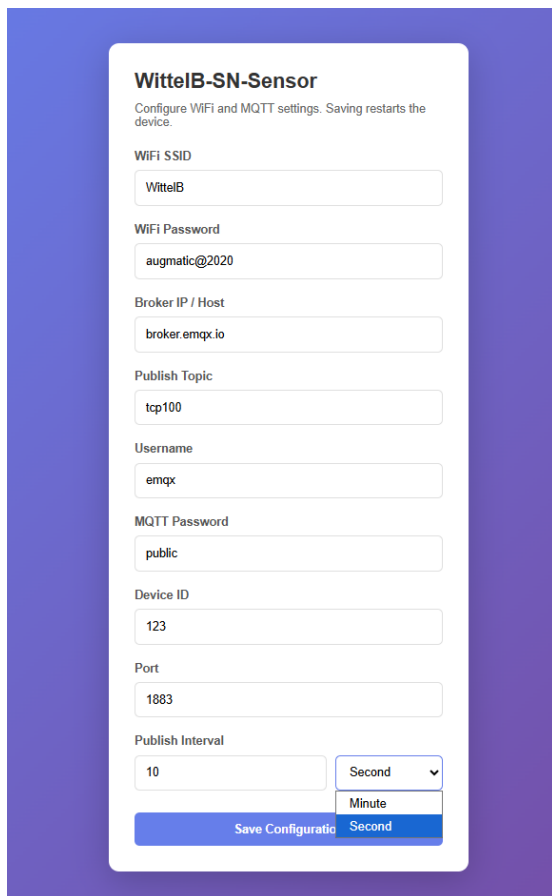


Image: 1



WittelB-SN-Sensor
 Configure WiFi and MQTT settings. Saving restarts the device.

WiFi SSID

WiFi Password

Broker IP / Host

Publish Topic

Username

MQTT Password

Device ID

Port

Publish Interval
 Second Minute

Save Configuration Second

Image: 2



wittelb 120

+ New Subscription JSON

Topic	QoS
tcp100	QoS 0
tcp10	QoS 0
tcp9	QoS 0
tcp101	QoS 0
tcp1	QoS 0
tcp15	QoS 0
tcp777	QoS 0

Topic: tcp100 QoS: 0

```
{
  "device": "123",
  "temp": "25.14",
  "hum": "43.96",
  "time": "10:55"
}
```

2026-06-26 10:55:52:676

Topic: tcp100 QoS: 0

```
{
  "device": "123",
  "temp": "25.17",
  "hum": "44.02",
  "time": "10:56"
}
```

2026-06-26 10:56:02:915

Topic: tcp100 QoS: 0

```
{
  "device": "123",
  "temp": "25.19",
  "hum": "44.09",
  "time": "10:56"
}
```

2026-06-26 10:56:12:639

Image: 3

- Enter IP 192.168.4.1 in web browser then you will see the Web UI.
- Enter the local WiFi credentials as shown in above image2 so the device can connect to that WiFi network. Using this connection, it will send the sensor data to the cloud.
- Then enter the required MQTT Credentials and make sure the credentials should be correct.
- As shown in Image 2, you can select the publish interval in either minutes or seconds. MQTT data will be published according to the selected interval.
- Enter the MQTT credentials to transmit the sensor data to the cloud.
- Image 3 shows the sensor data at cloud for the selected topic.

Contact Information

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